R Predictive Analytics Framework:
Malaysia Agriculture Yield

End-to-end framework that **analyses current crop yield patterns** and **predicts future yield production** to drive data-driven actionable insights.

WQD7004 *Programming For Data Science* | ***Prepared by:*** *Faiz Yah Bin Abdul Hanan Yah (24087980)*

1 The Problem

Current Challenge

Paddy Production Volatility
Inability to accurately predict the upcoming yield production

Yield optimisation
Identical farmland produces relatively inconsistent harvests

Consequences

Food Security Risks
Unpredictable rice threats nation self-sufficiency goals

Economic Losses
Farmers lacks accurate forecasting tools for resource allocation

Regional Disparities
Productivity gaps amongst regions remain unexplained

Investment Uncertainty
Agribusiness investors struggles with data-driven decision-making

2 The Dataset

Main Dataset

Department of Agriculture

Department of Statistic Malaysia (DOSM)

Crop Area and Production by State
- 6 years of data (2017-2022)
- 16 states
- 9 crop types

Link to Source: DOSM

Supplementary Dataset

Malaysian Meteorological Department (MET)

Weather dataset
- Data for temperature for each state
- Data for rainfall

Link to Source: MET

3 The Framework

Data Ingest
- Ingest crop production records from DOSM
- Integrate weather data via MET Malaysia API
- Merge datasets by state and date for unified analysis

Data Processing
- Handle missing values using state-specific median imputation
- Remove statistical outliers using IQR method
- Standardize production units to for all crop types

Feature Engineering
- Calculate productivity ratios
- Create year-over-year growth rate indicators
- Aggregate weather patterns by growing period

Model Prediction
- Train Random Forest and XGBoost regressors
- Implement 5-fold cross-validation to prevent overfitting
- Hyperparamater tuning using grid search to optimise model performance

Outcome and Insights
- Forecast upcoming crop yields with confidence intervals
- Identify top predictive features driving yield variation
- Build interactive Shiny dashboard for stakeholder exploration

R Libraries

readxl, readr	dplyr, tidyverse	caret, recipes, lubridate	randomForest, xgboost, caret	ggplot2, shiny

4 The Outcome

Quantifiable Benefits

1. **Improved forecasting accuracy**
   Farmers are more confident and knowing what to expect.
2. **Resources optimisation**
   Finance, workforce, time are organised to maximise efficiency.
3. **Early warning system**
   Stakeholders are notified beforehand to carry out necessary preparative measures.
4. **Data-drive policymaking**
   Policies are established based on evidence-backed insights to alleviate pain points.

Stakeholders Reaping the Benefits

Farmers

Farmer Cooperatives (FELDA, RISDA etc.)

Ministry of Agriculture and Food Security (MAFS)